

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 2

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Recap of Previous Lecture



Topic

Associative Memory

Topic

Locality of Reference

Topic

Cache Memory

Topics to be Covered



Topic

Cache Memory

Topic

Average Memory Access Time

Topic

Cache Write

#Q. Assume that for a certain processor, a read request takes 100 nanoseconds on a cache miss and 15 nanoseconds on a cache hit. Suppose while running a program, it was observed that 70% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is 40.5?

General formula:-

$$\begin{aligned} &= 0.7 * 15 + 0.3 * 100 \\ &= 40.5 \text{ ns} \end{aligned}$$

closer to CPU = cache

#Q. In a two-level hierarchy, if the top level has an access time of 10 ns and the bottom level has an access time of 60 ns, what is the hit rate on the top level required to give an average access time of 15ns?

far from CPU = mm

$$t_{cm} = 10 \text{ ns}$$

$$t_{mm} = 60 \text{ ns}$$

$$t_{avg} = 15 \text{ ns}$$

$$H = ?$$

Use hierarchical :-

$$15 = 10 + (1-H)60$$

$$5 = 60 - 60H$$

$$H = \frac{55}{60}$$

$$= 0.9167$$

$$= 91.67\%$$

#Q. In previous question if hit rate of the top-level memory is 100%, then the access time of bottom level memory will be 60 ns?

$$\begin{aligned} T_{avg} &= t_{cm} + 0 * t_{mm} \\ &= t_{cm} \end{aligned}$$



ans) $t_{cm} = 15 \text{ ns}$

$$t_{mm} = 100 \text{ ns}$$

hierarchical access

$$t_{avg} = 12 \text{ ns}$$

$$H = \text{---} \% \}$$

invalid question

$$\frac{t_{avg} < t_{cm}}{\downarrow}$$

not possible

#Q. Assume that for a certain processor, a read request takes 200 nanoseconds on a cache miss and 20 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The performance gain in memory access time of using cache as compared to not using cache in memory is _____?

⇒ w/o cache mem access time is 200 ns.

$$\begin{aligned}\text{with cache mem access time} &= t_{\text{avg}} = 0.8 * 20 + 0.2 * 200 \\ &= 16 + 40 \\ &= 56 \text{ ns}\end{aligned}$$

$$\text{performance gain of using cache} = \frac{200 \text{ ns}}{56 \text{ ns}} = 3.5714$$

In general :-

$$\text{Performance gain of using cache} = \frac{t_{mm}}{t_{avg}}$$

as compared to w/o cache

0.29 to 0.31

#Q. A computer system contains a cache. Uncached memory access takes 9 times longer than access to cache. If cache has a hit ratio 0.8. The ratio of cached memory access time to uncached memory access time is? (rounded to 2 decimal place)

Uncached memory $\Rightarrow$ only mm

Cached memory $\Rightarrow$ mem. system with cache $\Rightarrow$ avg

$$t_{cm} = x$$

$$t_{mm} = 9x$$

$$= \frac{t_{avg}}{t_{mm}} = \frac{0.8 * x + 0.2 * 9x}{9x} = \frac{2.6}{9} = \underline{\underline{0.29}}$$

$$\text{or} = \frac{x + 0.2 * 9x}{9x} = \underline{\underline{0.31}}$$



Topic : T_{avg} When Locality of Reference is Used

Sim.

$$t_{avg} = H * t_{cm} + (1-H) * t_{bt}$$

Hint:-

$$\begin{aligned} t_{avg} &= H * t_{cm} + (1-H) [t_{cm} + t_{bt}] \\ &= t_{cm} + (1-H) t_{bt} \end{aligned}$$

where t_{bt} = time needed to transfer a block from mm to cm

#Q. In a two-level hierarchy, the top level has an access time of 10 ns and hit rate of 90%. If the block transfer from main memory to cache takes 500ns in case of miss then average memory access time is 60?

$$\begin{aligned}t_{cm} &= 10\text{ns} \\ H &= 90\% \\ t_{bt} &= 500\text{ns}\end{aligned}$$

use hierarchical :-

$$= 10 + 0.1 * 500$$

$$= 60\text{ns}$$

#Q. Byte transfer time from mm to cache = 50ns

block size = 16 bytes

Block transfer time from mm to cache = 800 ns?

$$\begin{aligned} t_{bt} &= 16 * 50 \\ &= 800ns \end{aligned}$$



#Q. Byte transfer time from mm to cache = 50ns

block size = 16 words = $16 * 4 = 64$ bytes

1 Word = 4 bytes

Block transfer time from mm to cache = 3200 ns?

$$t_{bt} = 64 * 50 = 3200 \text{ ns}$$

- #Q. First byte transfer time from mm to cache = 10ns
Remaining each byte transfer time from mm to cache = 2ns
block size = 32 bytes
Block transfer time from mm to cache = 72 ns?

$$10 + (31 * 2) = 72 \text{ ns}$$

#Q. A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is _____?

$$\begin{aligned}
 \text{block size} &= \frac{256 \text{ B}}{64 \text{ bits}} \\
 &= \frac{256 \text{ B}}{8 \text{ B}} \\
 &= 32 \text{ words}
 \end{aligned}$$

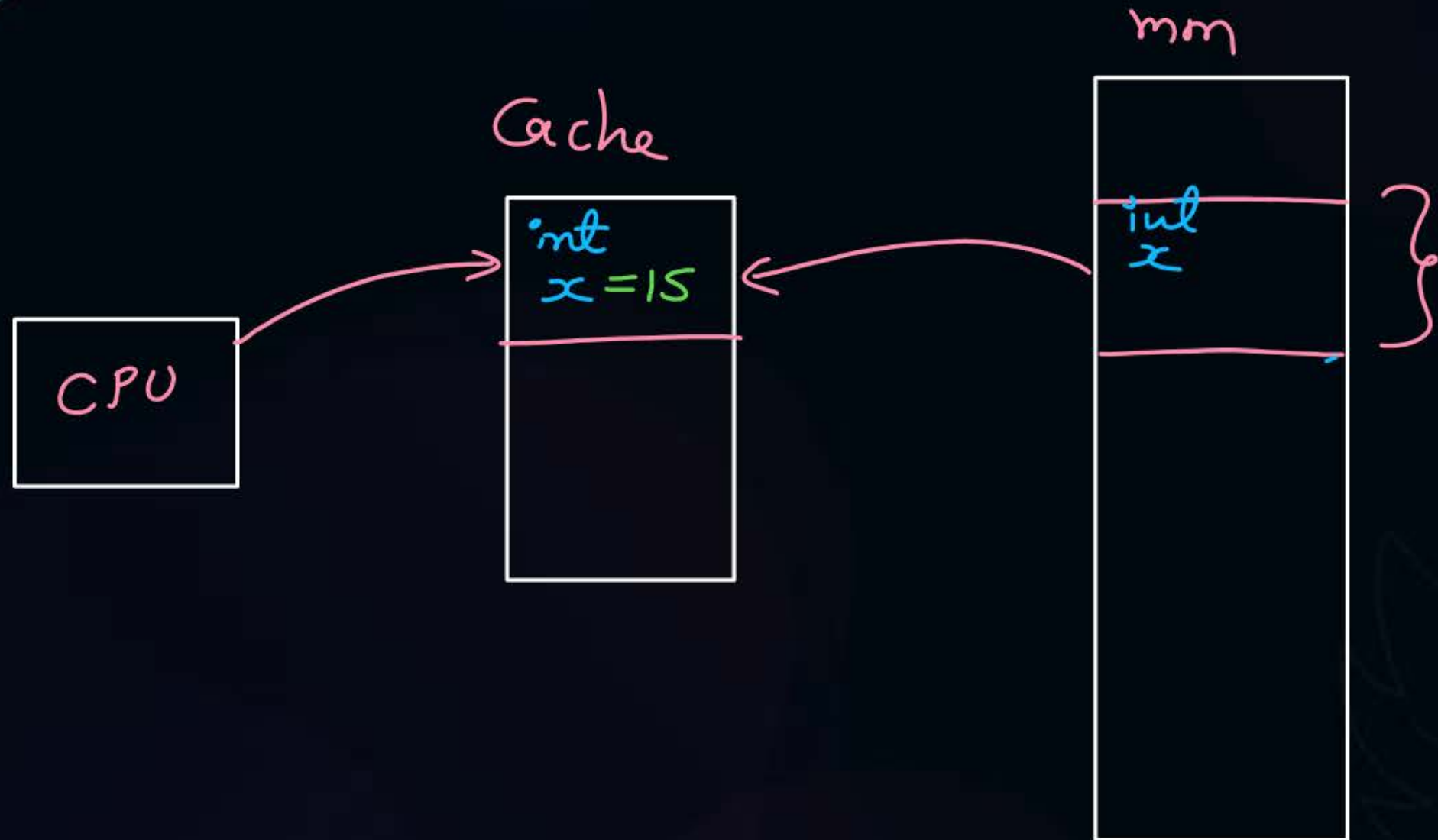
$$\begin{aligned}
 t_{bt} &= 20 + 31 * 5 \text{ ns} \\
 &= 175 \text{ ns}
 \end{aligned}$$

$$t_{avg}(\text{sim}) = 0.94 * 3 + 0.06 * 175 = 13.3 \text{ ns}$$

$$t_{avg}(\text{hier}) = 3 + 0.06 * 175 = 13.5 \text{ ns}$$



Topic : Cache Write or Write Propagation



if CPU performs write
in cache then mm
content also should
be updated.



Topic : Cache Write or Write Propagation

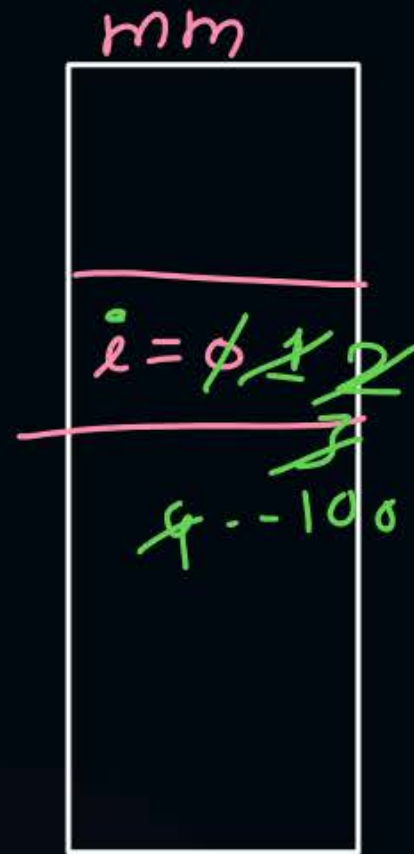
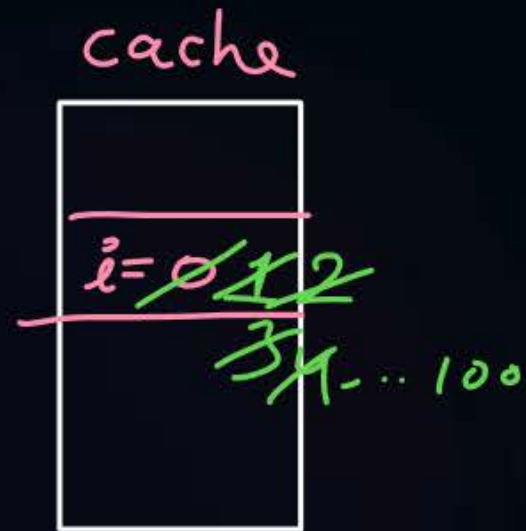
1. Write Through → uses simultaneous access by default
2. Write Back → uses hierarchical —||—



Topic : Write Through

CPU performs write in cache and mm simultaneously.

CPU



ex:-

```
for (i = 0; i < 100; i++)  
{  
}
```

advantage:- 1. Consistency in cache & mm content.

Disadvantage:- Time consuming because for write operation mm is accessed even if there is cache hit.

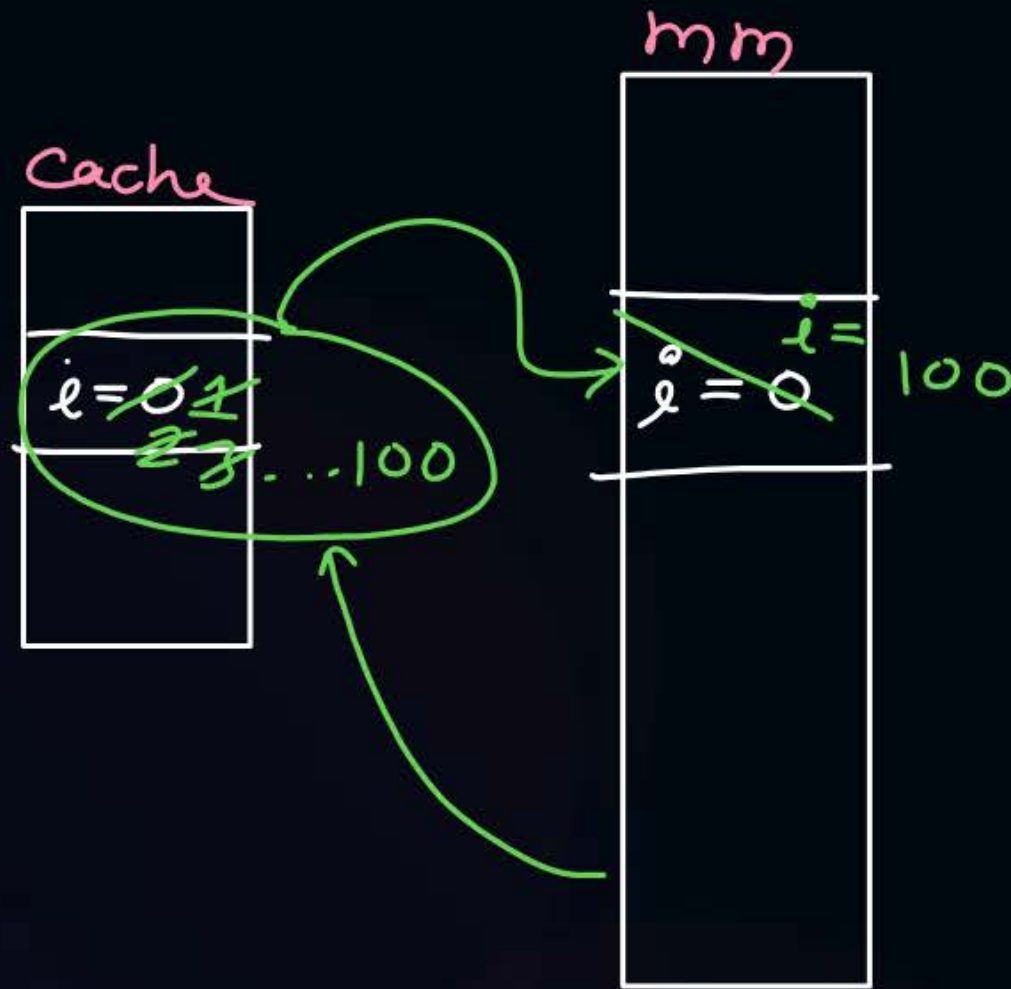


Topic : Write Back

CPU performs write only in cache and the block is written back to mm when it is replaced.



CPU



advantage:-

→ Time saving
(access cache only for hit)

Disadvantage:-

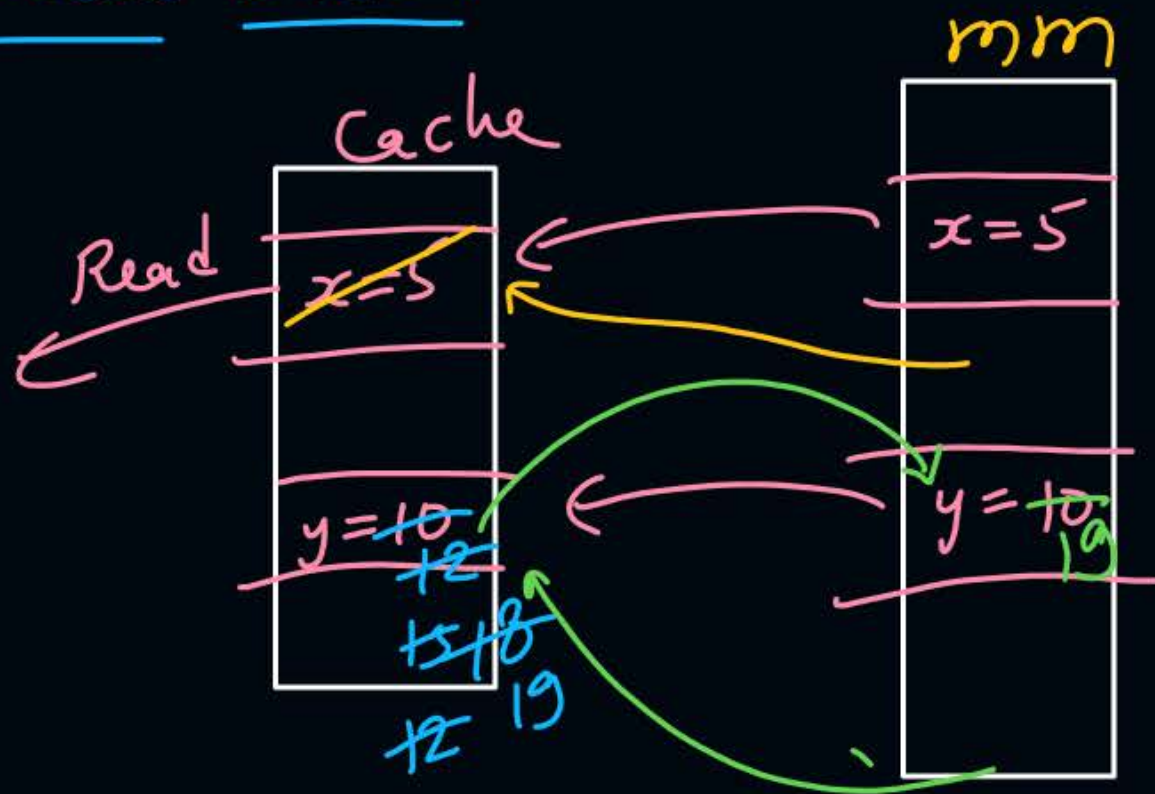
→ Inconsistency in values of cache and mm.

Block replacement

write through:-

when a block is replaced from cache, then there is no any need of write back of replaced block ever.

write back:-



when a block is replaced from cache then it will need write back to mm only if CPU has performed write into it.

The blocks in which CPU performed write are called as dirty or modified blocks.

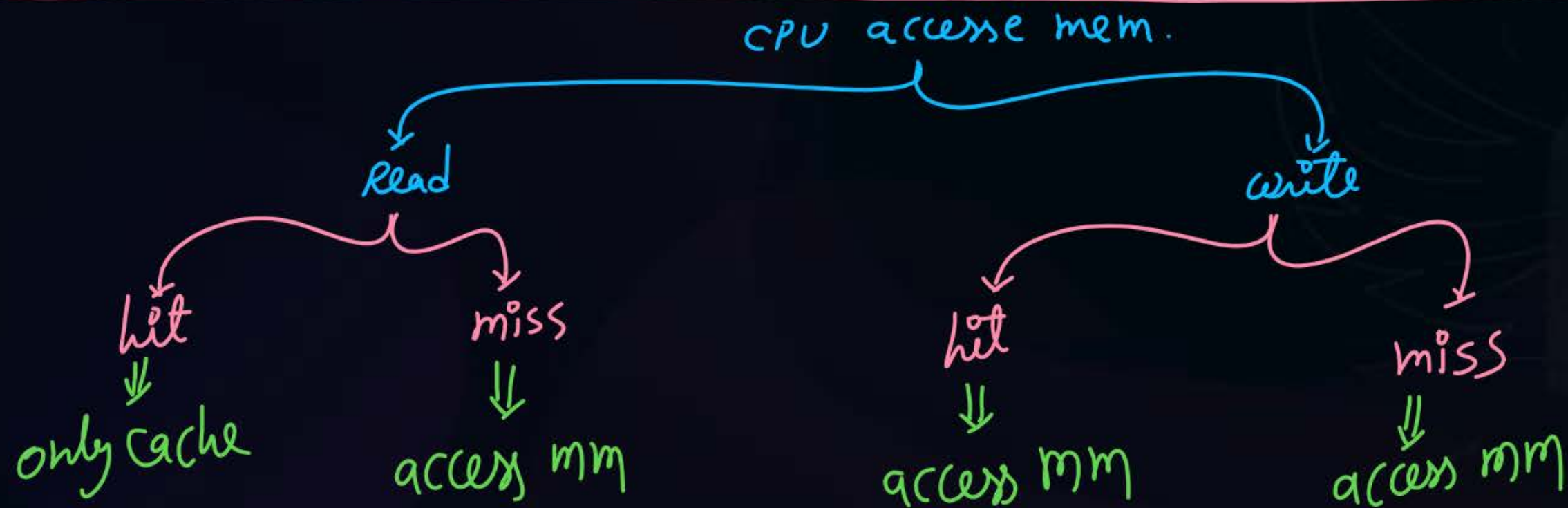


Topic : T_{avg} in Write Through Cache

$$t_{avg \text{ read}} = H * t_{cm} + (1-H) t_{mm}$$

$$t_{avg \text{ write}} = t_{mm}$$

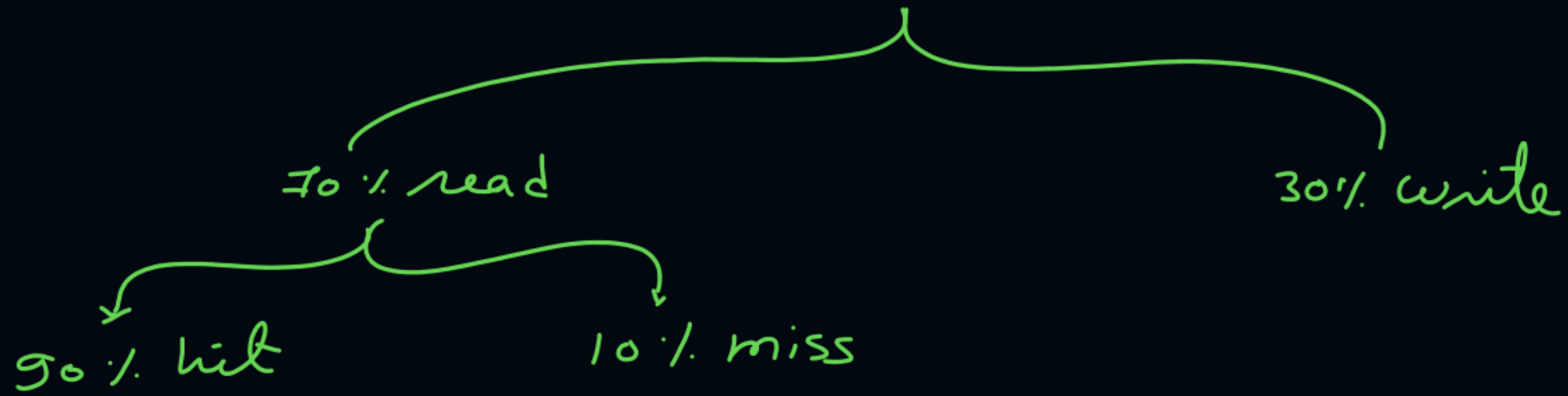
$$t_{avg} = \text{fraction of read operations} * t_{avg \text{ read}} + \text{fraction of write operations} * t_{avg \text{ write}}$$



$$\text{effective hit rate of cache in write through} = \text{fraction of read operations} * \text{hit ratio}$$

#Q. A system has a write through cache with access time of 100ns and hit ratio of 90%. The main memory access time is 1000ns. The 70% of memory references are for read operations.

- A** Average memory access time for read operations only = $0.9 * 100 + 0.1 * 1000 = 190 \text{ ns}$
- B** Average memory access time for write operations only = 1000 ns
- C** Average memory access time for read-write operations both = $0.7 * 190 \text{ ns} + 0.3 * 1000 = 433 \text{ ns}$
- D** Effective Hit ratio = $0.7 * 0.9 = 0.63$ or 63%



$$\text{Effective hit rate} = 0.7 * 0.9 \\ = 0.63$$

or

63%



2 mins Summary



Topic

Cache Memory

Topic

Average Memory Access Time

Topic

Cache Write



Happy Learning

THANK - YOU

